LESSON 3.4

Determine Equivalent Ratios Using Graphs

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

WHY THIS MATTERS The organizers of a soccer league ordered 6 of the bags shown to hold soccer balls. Each bag holds the same number of soccer balls. How many soccer balls will the bags hold?

REMEMBER

Determine Equivalent Ratios Using Graphs

Graphing Equivalent Ratios An ordered pair (x, y) tells where to plot a point: go right x, then up y. When you graph equal ratios, the points make a straight line through the origin (0, 0).

1. Form ordered pairs (x, y) from columns or rows in a ratio table.
2. Plot the points on the coordinate plane.
3. Proportional Relationship: Points form a straight line passing through the origin (0, 0)

Words: **Coordinate plane *(Plano cartesiano)*** **Ordered pair *(Par ordenado)*** **Linear pattern *(Patrón lineal)*** **Proportional *(Proporcional)*** **Origin *(Origen)***

1 **Circle the letter of the best answer. Show how you know.**

A ratio table for 2:5 starts with 2:5, 4:10. What is the third row?

(A) 5:10
(B) 6:12
(C) 6:15
(D) 8:15

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which row does NOT belong in a ratio table for 4:5?

(A) 6:10
(B) 8:10
(C) 12:15
(D) 16:20

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A ratio table shows 3:7, 6:14, and 9:21. What is the next row?

(A) 10:24
(B) 12:24
(C) 12:28
(D) 15:28

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A recipe uses 3 eggs for every 5 cups of flour. How many eggs are needed for 15 cups of flour?

(A) 6
(B) 9
(C) 10
(D) 12

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos. The graph shows 1 taco uses 2 oz of salsa. How much salsa for 15 tacos?

(A) 7.5 oz
(B) 17 oz
(C) 20 oz
(D) 30 oz

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos. Why do the points form a straight line through (0, 0)?

(A) Because the quantities are proportional — the same ratio holds for every pair
(B) Because the owner sold the same number of tacos every day
(C) Because all graphs of data are straight
(D) Because salsa is measured in ounces

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

SENTENCE STARTER Every time the sundaes go up by ___, the ounces go up by ___, so the points line up in a ___ starting from ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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Set B · Version B · Second form · core practice

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LEARNING TARGET I can graph the values from a ratio table as points on the coordinate plane.

WHY THIS MATTERS The organizers of a soccer league ordered 6 of the bags shown to hold soccer balls. Each bag holds the same number of soccer balls. How many soccer balls will the bags hold?

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each cups-to-scoops table row to the ordered pair you would plot (rule: y = 2x).

_____ 2 cups → 4 scoops

_____ 3 cups → 6 scoops

_____ 5 cups → 10 scoops

_____ 6 cups → 12 scoops

(A) (2, 4)

(B) (3, 6)

(C) (5, 10)

(D) (6, 12)

2 **Write the name of the correct group on each line.**

The smoothie line follows the rule y = 2x (2 scoops per cup). Sort each ordered pair: is it ON the line or OFF the line?

Groups: ON the line (y = 2x) | OFF the line

_____ (1, 2)

_____ (4, 8)

_____ (5, 10)

_____ (2, 3)

_____ (3, 5)

_____ (1, 4)

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos. Which point would NOT be on this line?

(A) (4, 8)
(B) (5, 12)
(C) (6, 12)
(D) (10, 20)

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Recipe: 1 cup berries to 2 scoops yogurt. With cups on x and scoops on y, which pair shows 3 cups?

(A) (3, 6)
(B) (3, 5)
(C) (3, 3)
(D) (6, 3)

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

On the smoothie graph, every point follows the rule scoops = 2 × cups. Which equation describes the pattern between x (cups) and y (scoops)?

(A) $y = 2x$
(B) $y = x + 2$
(C) $y = x \div 2$
(D) $x = 2y$

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — The Swapped Smoothie Point

1. **Read the table** Cups of blueberries: 2 | Scoops of yogurt: 4 (rule: 2 scoops per cup)
2. **Name the axes** Cups go on the x-axis, scoops go on the y-axis
3. **Write the ordered pair** (4, 2)
4. **Plot the point** Go right 4, up 2 on the grid

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What pattern do the plotted points make? How does the graph show the ratio of 1 sundae to 2 ounces of sauce?

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